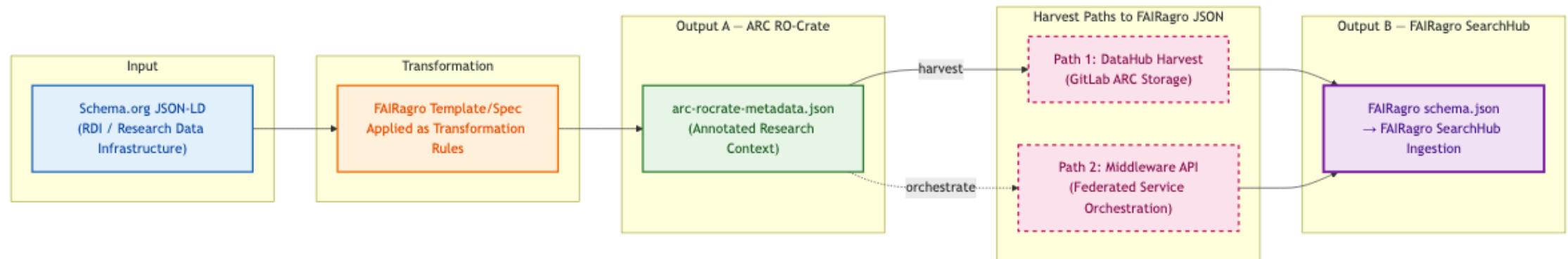


FAIRweaver: Schema.org → ARC → FAIRagro Workflow

Slide 1 — FAIRagro Metadata Transformation Pipeline



Slide 1 — Key Points

- **Sequential dependency:** Output B (FAIRagro JSON) is derived *from* Output A (ARC RO-Crate) — not a parallel output
- **Two harvest paths** converge on the same FAIRagro JSON schema:
 - **Path 1 (solid):** Direct harvest from GitLab DataHub where ARCs are stored
 - **Path 2 (dashed):** Via FAIRagro Middleware API (federated service that orchestrates the full workflow)
- Both paths produce identical `FAIRagro schema.json` ingested into SearchHub

Slide 2 — Three File Scenarios: Input → ARC → FAIRagro Output

Case	Input File	ARC Output	FAIRagro Output
Synthetic	schema-org-wheat-full.json	arc-ro-crate-wheat-full ✓ compliant	Full extraction ✓
Real — Small	arc-ro-crate-dronflyover.json (<10 MB)	Manual, partial ⚠	Partial — mappable fields only
Real — Large	arc-ro-crate-muenchenberg-lte.json (>100 MB)	Manual, partial ⚠	Basic harvest only

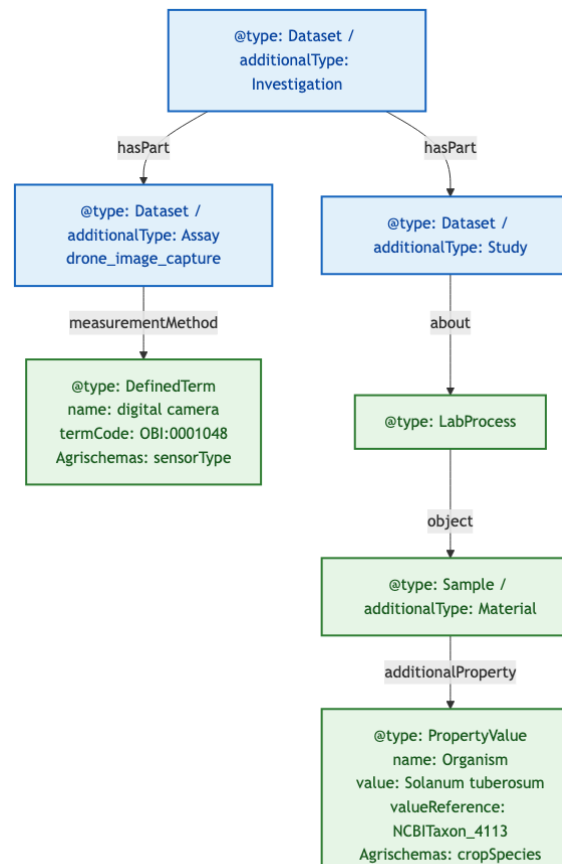


If an ARC follows the FAIRagro specification → full metadata extraction. If not → only basic information is harvested.

Slide 3 — Examining ARC Structure: Domain Objects at Different Depths

Goal:

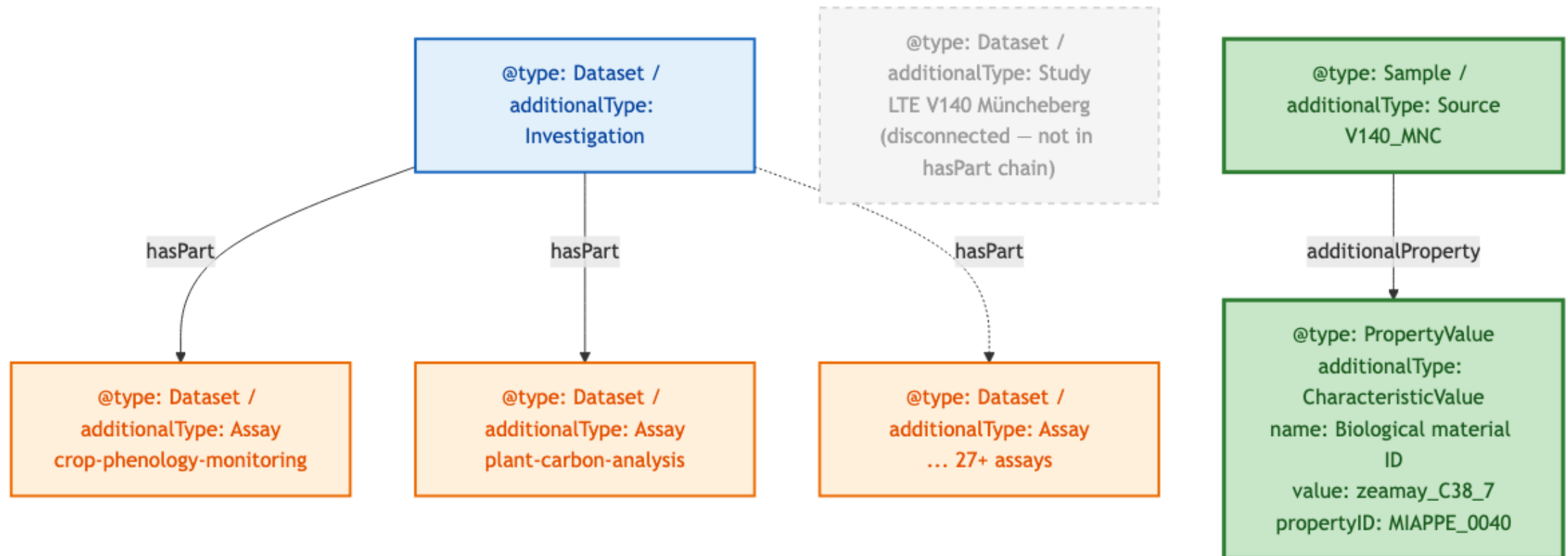
- Understand how Agrischemas concepts map into ARC RO-Crate
- Show that equivalent domain concepts require very different traversal depths



Slide 4 — Müncheberg ARC: A Different Structural Pattern

Goal:

- Show another real ARC with a different structural pattern
- Reinforce that parser must handle multiple modeling conventions



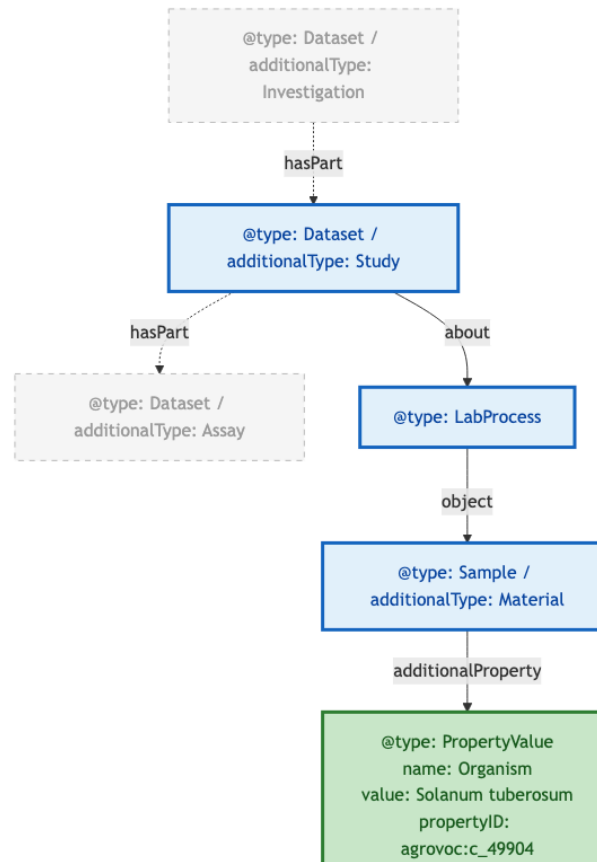
Slide 4 — Comparison: Drone Flyover vs Müncheberg

Aspect	Drone Flyover	Müncheberg LTE
Study entity	Explicit, in hasPart chain	Present but disconnected (not in hasPart)
Crop species path (short)	Study → LabProcess → Sample → PropertyValue (4 hops)	Source → additionalProperty → CharacteristicValue (2 hops)
Crop species path (long)	Same as short (only path)	ALSO via Study/LabProcess → object → Source → additionalProperty
Sensor metadata	Present (DefinedTerm)	Absent
Assay count	1	27+

Slide 5 — Required Modeling Pattern & Standardization Gap

Goal:

- Define the required path for unambiguous extraction
- Identify what still needs standardization



Slide 5 — Open Questions

Structure: ?	How to formally specify the required traversal path?
propertyID: SSSOM mapping	How to standardize ontology term mappings?